STUDYNEST | CLASS 11 PHYSICS

MARKING SCHEME AND MODEL ANSWERS

Chapter 14: Waves

Maximum Marks: 40 | Suggested time: 60 minutes

Use after attempting the paper. Equivalent correct methods should receive credit.

Q1 1 mark

Perpendicular.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q2 1 mark

Longitudinal.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q3 1 mark

v=flambda.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q4 1 mark

2pi/lambda.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q5 1 mark

Resultant displacement is the algebraic sum.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q6 1 mark

Pulse is inverted.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q7 1 mark

|f1-f2|.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q8 1 mark

Maximum displacement from equilibrium.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q9 2 marks

100 m/s.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q10 2 marks

2 m.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q11 2 marks

4 Hz.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q12 2 marks

Doubles.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q13 2 marks

No.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q14 2 marks

+x direction.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q15 3 marks

30Hz; 1/30s.

Suggested split: 1 for the relation, 1 for working/reasoning and 1 for result with unit. Accept an equivalent correct method.

Q16 3 marks

+2 cm.

Suggested split: 1 for the relation, 1 for working/reasoning and 1 for result with unit. Accept an equivalent correct method.

Q17 3 marks

4 Hz; .25 s.

Suggested split: 1 for the relation, 1 for working/reasoning and 1 for result with unit. Accept an equivalent correct method.

Q18 3 marks

Particles oscillate locally while the disturbance propagates.

Suggested split: 1 for the relation, 1 for working/reasoning and 1 for result with unit. Accept an equivalent correct method.

Q19 4 marks

30m/s; lambda=6m.

Suggested split: 1 for principle, 1 for equation, 1 for working and 1 for result/explanation. Accept an equivalent correct method.

Q20 4 marks

Alternating loudness occurs at 2 beats per second.

Suggested split: 1 for principle, 1 for equation, 1 for working and 1 for result/explanation. Accept an equivalent correct method.